

LPAR Design Tool

*LPAR weights. Defined Capacities. Capacity Group Limits. HiperDispatch. Vertical Highs, Mediums, and Lows. These are just some of the considerations when optimizing your LPAR configuration. But they are enough to make your head spin. Luckily for us, **Alain Maneville**, from IBM France, appears to be impervious to dizziness. And to help the rest of us, Alain created a tool called LPAR Design.*



Alain Maneville

You might remember that Alain (the handsome, debonair, young chap in the photo) worked with us on the ‘[HiperDispatch Questions and Answers](#)’ article in [Tuning Letter 2015 No. 4](#).

Alain and one of his customers presented an excellent session at the zUniversity in Munich this year about using the LPAR Design tool together with IBM’s zPCR tool to prepare for an upgrade from a z13 to a z14. Like the zPCR tool, there is no charge for the LPAR Design tool.

We have been hoping for many months to have a chance to work with Alain on an article about his LPAR Design tool. With the recent announcement of the IBM z14, and the ‘[CPU MF Part 3 - Optimizing CPC Cache](#)’ article in this issue, this seemed like the perfect time to introduce our readers to this very useful tool. We were also very fortunate to have the help of ‘a large European customer’, who kindly provided the SMF data that allowed us to illustrate the use of LPAR Design in a real-world, complex, environment.

We hope that you will enjoy the article and will make the LPAR Design tool a standard part of your performance analyst toolkit. If you have any questions, or suggestions for future enhancements to the LPAR Design tool, please email us at technical@watsonwalker.com, or Alain directly, at alain_maneville@fr.ibm.com.

Introduction

LPAR weights are used to indicate the share of the CPC’s capacity that an LPAR is guaranteed if every LPAR on the CPC is trying to get as much capacity as possible. For this reason, an LPAR’s weight is also sometimes referred to as its ‘fair share’. An LPAR’s relative weight is also a key part of the HiperDispatch logic for deciding how many Vertical High, Medium, and Low CPs the LPAR will have. A tiny difference in an LPAR’s weight can increase or decrease the number of Vertical High CPs in the LPAR. For these reasons, it is vital that you understand the relationship between your LPAR definitions and the resulting HiperDispatch topology.

Let's look at an example. Let's say that you have 3 LPARs; WAWA, WAWB, and WAWC; with weights of 500, 300, and 200 respectively. The sum of the LPAR weights is 1000, so WAWA's fair share is 500/1000 (50%), WAWB's is 300/1000 (30%), and WAWC's is 200/1000 (20%). If these LPARs are running on a 10-way CPCs, WAWA would be entitled to 5 CPs, WAWB would be entitled to 3, and WAWC would be entitled to 2. Simple.

But remember that the weights are *relative* values. Specifically, they are relative to the sum of the weights of all *active* LPARs. So, what would happen if you add a new LPAR, WAWD, to the configuration? The fair share of the three existing LPARs will change, depending on the weight you assign to WAWD.

Alternatively, what would happen if WAWA's weight was increased to 600? You might think that would give it a fair share of 60% of the available capacity. However, its fair share is now 600/1100, or 54.5% - remember that every weight change potentially alters the denominator that is used to calculate the fair share for every LPAR.

None of these examples are very complex. However, what if the CPC has 30 LPARs instead of 3? And 29 CPs, with some of them being dedicated rather than shared, and if the number of shared CPs doesn't easily divide into the summed weights of all active LPARs? And don't forget that LPARs also have a separate set of weights to determine how much zIIP capacity they are guaranteed.

How do you easily ensure that every LPAR is defined with enough logical CPs to be able to get all of the capacity 'guaranteed' by its relative weight? And how will your relative weights translate into Vertical High, Medium, and Low CPs? And if the weight of an LPAR guarantees it 1.6 CP's worth of capacity, will the LPAR have 1 VH CP and 1 VM CP with a 60% share? Or will it have 2 VM CPs, each with an 80% share?

The answers to all these questions can be calculated, but it is a time consuming chore. And you need to be sure that you are applying the correct HiperDispatch rules for *your* CPC (because IBM *does* change the rules from time to time).

Alain originally created his LPAR Design spreadsheet to help his customers easily answer these questions. As the tool grew in popularity, it naturally also grew in function, with the latest iteration being Version 9.2. This article illustrates how you can use the tool to help manage and optimize your LPAR topology. To get the maximum value from this article, it should be used in conjunction with the '[HiperDispatch Questions and Answers](#)' article in [Tuning Letter 2015 No. 4](#) and the '[CPU MF Part 3 - Optimizing CPC Cache](#)' article in this issue. You should also review **Kathy Walsh's** excellent TechDoc [106388](#), *Number of Logical CPs Defined for an LPAR*, before making any adjustments to your LPAR topology.

What Can I Do With the LPAR Design Tool?

Level Set

The LPAR Design tool analyzes *the defined configuration*. It is *not* a tool for analyzing utilization or workload RNIs. Its objective is to help you understand how HiperDispatch interprets your LPAR definitions and maps them to Vertical High, Medium, and Low logical processors. It can help you better understand your existing configuration, and prepare for changing LPAR weights, adding or removing LPARs, or migrating to a new or upgraded CPC.

Having used the tool to fine tune your configuration, you can then export that configuration back to zPCR to make sure that your new configuration can handle the workload's capacity needs.

Before we start getting into detail about what the tool can do, it would probably be a good idea for you to download the latest version so that you can navigate around the tool as we describe its different parts. The LPAR Design tool is available from the WLM home page, at https://www.ibm.com/systems/z/os/zos/features/wlm/WLM_Further_Info_Tools.html#Design. The download consists of a single zip file that contains the User's Guide in PDF format, and the LPAR Design spreadsheet (a .xls file).

Having downloaded the tool, what can you do with it? The first thing you will want is to define your configuration to the tool. You can achieve this in one of two ways:

- ◆ Manually, using facilities in the tool.
- ◆ In an automated way, using .zpcr files created with the IBM zPCR tool¹.

We recommend this method, especially if you have a larger, more complex, configuration.

The provided .xls file comes with some sample data already loaded into it. If you wish, you can use that sample data at this point, and then load your own data later (as described in ['Defining Your Configuration to LPAR Design' on page 22](#)).

IMPORTANT LESSON: *Never* manually add or remove rows or columns to any of the worksheets in the tool. Doing so will inevitably result in one of those ever-popular Microsoft Visual Basic runtime error messages.

If you need to delete an LPAR from the configuration, use the 'Delete selected LPAR' button on the CONFIG and CONFIG-IFL sheets.

As you will see, the LPAR Design tool consists of a number of worksheets.

¹ zPCR can be downloaded (at no charge) from <https://www.ibm.com/support/techdocs/atsmastr.nsf/WebIndex/PRS1381>

The START sheet is presented every time you open an LPAR Design spreadsheet. Once you have loaded or created your configuration, you should only need to return to this sheet if you want to change the CPC type or model (if you are modeling a different configuration, for example).

There are separate sheets for each of general purpose CPs, zIIPs², and IFLs (called CONFIG, CONFIG-zXXP, and CONFIG-IFL respectively). The worksheets show the number of Vertical High, Medium, and Low CPs, zIIPs, and/or IFLs for each LPAR. They also show the percent of an engine that is shared by the LPAR's Vertical Medium(s). For example, if an LPAR has a fair share of 2.89 CPs, the CONFIG sheet would show that it has 2 Vertical Highs, and 1 Vertical Medium, with the Medium having an 89.1% share. The sheet also draws attention to attributes that don't conform to IBM's CPC best practice guidelines - for example, if an LPAR has more than two Vertical Low CPs.

Why Limit the Number of Vertical Low CPs?

When deciding which logical CP to dispatch on a physical CP, PR/SM checks the recent CPU consumption of that logical CP and compares that to its fair share. The fair share of a Vertical Low CP is equal to the share of the LPAR's VM CP(s) divided by the number of VMs plus the number of unparked VLs. As WLM unparks more VLs, the fair share of each one decreases. As a result, the chance of those VLs being dispatched by PR/SM also decreases.

That in turn can result in each VL getting less capacity, leading to WLM unparking more VLs. We know that VL CPs nearly always have a poorer RNI than VH CPs, so the more unparked CPs you have, the higher is the chance of any unit of work ending up on a VL CP. Also, there are typically more VL CPs in an affinity node containing VL CPs than there are in an affinity node containing only VH CPs. Because SRM queues work at the affinity node level, the more logical CPs there are in an affinity node, the lower is the chance of work being re-dispatched on the same logical CP that it was running on previously. Also, WLM, by design, does not make any attempt to limit high importance work to VH and/or VM CPs, so the more unparked VL CPs you have, the greater is the chance of high importance work ending up on a VL CP, with all the negative performance implications that entails.

For more discussion about the desired number of VL CPs in an LPAR, refer to the 'Q15: Tell me more about VLs. Are they good? Bad? How many should I have?' section in the ['HiperDispatch Questions and Answers'](#) article in *Tuning Letter 2015 No. 4*.

In addition to showing information about the current configuration, you can also use these sheets to model the impact of making a change to the configuration. For example, you can

² The current version of the LPAR Design tool, V9, no longer supports zAAPs. If you have an older CPC that has zAAPs and you want to use LPAR Design, email Alain and he will send you an old version of the tool that still supports zAAPs.

determine the impact of changing the weight of one (or more) LPARs on the mix of Vertical High, Medium, and Low CPs in every LPAR. Or, you can see how moving to a sub-capacity model with more CPs would affect the mix of VH, VM, and VL CPs. Or, if you have carefully planned your weights to optimize the number of VH CPs, you can see what adjustments you would need to make to ensure that your optimization is carried forward to a new CPC.

The CONFIG sheet is shown in Figure 3. You can override any of the values in the yellow columns (LPAR weight, number of logical CPs, and the workload LSPR RNI category) to model the impact of those changes³. If you want to model a change to the underlying CPC, however, you would make that change on the START sheet. The LPAR Design User's Guide contains a description of all the fields in the worksheet.

TIP: Any time you make a change to the configuration, you have to click on the Configuration Validation button to apply those changes.

Figure 3 - CONFIG Worksheet

ID= - LPARDesign-HD-V9-T02 Current zPCR Version-9.1a - SpecCtg=YES LPAR DEFINITION (CP)														
CFG-UP-VALID?	YES		#PhyProc	36	Shared-Pool									
Machine-type	2964-736		#LPs (non-ICF, non-DED)	254	Ded-Pool									
MSU	4,452		Ratio LP/PP (base)	7.06	0									
Total Weight	421		LSPR-AVG-V2R2-MI	37,972										
Max LPAR	85		#LPARs	23										
HD supported on 2964														
LPARNAME	WEIGHT	#LP	%SHARE (By Pool)	RESERVED	Guaranteed #PP	Wkld LSPR	MinReq#LP	Check#LP	HD-HIGH#	HD-MED#	HD-MED%	HD-LOW#	Active LP	Report LP
WAW0B1	173	30	41.1%		14.79	Average	15	#VL>2	14	1	79.3%	15	15	15
WAW0T0	1	2	0.2%		0.09	Average	1	OK	0	1	8.6%	1	2	1
WAW0T1	4	2	1.0%		0.34	Average	1	OK	0	1	34.2%	1	2	1
WAW0X1	11	10	2.6%		0.94	Average	1	#VL>2	0	1	94.1%	9	2	1
WAW0X5	8	5	1.9%		0.68	Average	1	#VL>2	0	1	68.4%	4	2	1
WAW0Y1	10	10	2.4%		0.86	Average	1	#VL>2	0	1	85.5%	9	2	1
WAW0Y5	3	5	0.7%		0.26	Average	1	#VL>2	0	1	25.7%	4	2	1
WAW0Z0	2	2	0.5%		0.17	Average	1	OK	0	1	17.1%	1	2	1
WAW0Z1	44	20	10.5%		3.76	Average	4	#VL>2	3	1	76.2%	16	4	4
WAW0Z5	10	5	2.4%		0.86	Average	1	#VL>2	0	1	85.5%	4	2	1
WAW0B0	12	4	2.9%		1.03	Average	2	OK	0	2	51.3%	2	2	2
WAW0B3	11	25	2.6%		0.94	Average	1	#VL>2	0	1	94.1%	24	2	1
WAW0B5	36	25	8.6%		3.08	Average	4	#VL>2	2	2	53.9%	21	4	4
WAW0B7	14	30	3.3%		1.20	Average	2	#VL>2	0	2	59.9%	28	2	2
WAW0D0	4	2	1.0%		0.34	Average	1	OK	0	1	34.2%	1	2	1
WAW0D1	11	10	2.6%		0.94	Average	1	#VL>2	0	1	94.1%	9	2	1
WAW0Q1	10	10	2.4%		0.86	Average	1	#VL>2	0	1	85.5%	9	2	1
WAW0Q5	4	5	1.0%		0.34	Average	1	#VL>2	0	1	34.2%	4	2	1
WAW0R0	2	2	0.5%		0.17	Average	1	OK	0	1	17.1%	1	2	1
WAW0R1	29	20	6.9%		2.48	Average	3	#VL>2	1	2	74.0%	17	3	3
WAW0R3	8	5	1.9%		0.68	Average	1	#VL>2	0	1	68.4%	4	2	1
WAW0R5	7	10	1.7%		0.60	Average	1	#VL>2	0	1	59.9%	9	2	1
WAW0R7	7	15	1.7%		0.60	Average	1	#VL>2	0	1	59.9%	14	2	1

Figure 3 shows a sample CONFIG worksheet. This example is from a z13 736 with 23 z/OS LPARs. The CONFIG-zXXP and CONFIG-IFL worksheets contain pretty much the same information, but for zIIPs and IFLs.

The next worksheet is the CONFIG-MSU one. This worksheet lets you define your LPAR and Group caps, and see how those caps could affect the capacity that is available to each LPAR. I'm sure that many of our readers know **Robert Vaupel** from the WLM development group in Boeblingen; Robert was the one that created the code for the CONFIG-MSU sheet.

³ The Workload Characterization information is not used by the LPAR Design tool itself. It is only held so that it can be inserted into the .zpcr file if you use LPAR Design to create one.

Nearly every installation today uses some form of capping. Capping can be a very effective way of controlling your software costs. However, when you consider the plethora of capping options that are available, and the fact that multiple options can be used for the same LPAR at the same time, trying to determine precisely how much capacity an LPAR can have when it is capped can be quite a challenge. The CONFIG-MSU worksheet is provided to help you address that challenge.

IMPORTANT: The mixture of Vertical High, Medium, and Low CPs in an LPAR is impacted by the LPARs weight *and*, potentially, by an LPAR being capped. However, soft capping only limits the LPAR's capacity when the Rolling 4-Hour Average exceeds the cap value. In our experience, most LPARs run uncapped for most of the time. **Therefore, all the worksheets in LPAR Design, except for the CONFIG-MSU sheet, ignore any caps that might be defined for the LPARs.**

Figure 4 - CONFIG-MSU Worksheet

A

B

C

D

E

F

G

H

I

J

K

L

M

N

O

P

Q

R

ID-Watson & Walker - LPARDesign-HD-V5-T02

Current zPCR Version 3.1a - SpecCtg=YES

Group Capping

Minimize/Maximize

CEC

2964.736

PCPs

36

MSU

4,452

Total Weight

421

Shared Pool

36

Clear Definitions

Clear Calculations

Calculate

Definitions						CEC/LCP based			Group Calculations								Result	
Lpars	LCPs	Weight	Defined Capacity Limit	Capacity Group Name	Capacity Group Limit [MSU]	Share [%]	MSU at Weight	Theoretical Usable MSU	Total Group Weight	Group Share [%]	Group Share [MSU]	Possibly Donated MSU	Total Donated MSU	Possible Group Receiver	Group Receiver Share [%]	Received Donated MSU	Maximum Consumable MSU	Comment
WAW007	30	14		DB2	410	3.3%	148.0	3,710.0	32	43.8%	179.4		24.7	YES	56.0%	13.8	193.2	cap pattern or negative phantom weight
WAW010	2	1		GP	100	0.2%	10.6	247.3	21	4.8%	4.8						4.8	positive phantom weight
WAW011	2	4	40	IMS	2,015	1.0%	42.3	247.3	281	1.4%	28.7						28.7	positive phantom weight
WAW001	10	11	140	IMS	2,015	2.6%	116.3	1,236.7	281	3.9%	78.9						78.9	positive phantom weight
WAW005	5	8		CICS	290	1.9%	84.6	618.3	68	11.8%	34.1						34.1	positive phantom weight
WAW011	10	10	90	IMS	2,015	2.4%	105.7	1,236.7	281	3.6%	71.7						71.7	positive phantom weight
WAW005	5	3		CICS	290	0.7%	31.7	618.3	68	4.4%	12.8						12.8	positive phantom weight
WAW020	2	2		GP	100	0.5%	21.1	247.3	21	9.5%	9.5						9.5	positive phantom weight
WAW021	20	44	380	IMS	2,015	10.5%	465.3	2,473.3	281	15.7%	315.5						315.5	positive phantom weight
WAW025	5	10		CICS	290	2.4%	105.7	618.3	68	14.7%	42.6						42.6	positive phantom weight
WAW080	4	12		GP	100	2.9%	126.9	484.7	21	57.1%	57.1						57.1	positive phantom weight
WAW081	30	173		IMS	2,015	41.1%	1,829.4	3,710.0	281	61.6%	1,240.6						1,240.6	positive phantom weight
WAW083	25	11		ZOS	80	2.6%	116.3	3,091.7	19	57.9%	46.3						46.3	positive phantom weight
WAW085	25	36		CICS	290	8.6%	380.7	3,091.7	68	55.9%	153.5						153.5	positive phantom weight
WAW002	7	4	100	GP	100	1.0%	42.3	247.3	21	10.0%	19.0						19.0	positive phantom weight
WAW001	10	11		DB2	410	2.6%	116.3	1,236.7	32	34.4%	140.9		24.7	YES	44.0%	10.9	151.8	cap pattern or negative phantom weight
WAW001	10	10	90	IMS	2,015	2.4%	105.7	1,236.7	281	3.6%	71.7						71.7	positive phantom weight
WAW005	5	4		CICS	290	1.0%	42.3	618.3	68	5.9%	17.1						17.1	positive phantom weight
WAW002	2	2		GP	100	0.5%	21.1	247.3	21	9.5%	9.5						9.5	positive phantom weight
WAW001	20	29	370	IMS	2,015	6.9%	306.7	2,473.3	281	10.3%	208.0						208.0	positive phantom weight
WAW003	5	8	50	ZOS	80	1.9%	84.6	618.3	19	42.1%	33.7						33.7	positive phantom weight
WAW005	10	7		CICS	290	1.7%	74.0	1,236.7	68	10.3%	29.9						29.9	positive phantom weight
WAW007	15	7	65	DB2	410	1.7%	74.0	1,855.0	32	21.9%	88.7	24.7					65.0	positive phantom weight

START

CONFIG

CONFIG-MSU

CONFIG-MSU-2

CONFIG-PL

EXPERT

SYNTHESIS

DASHBOARD HD

SINET

Tables

The CONFIG-MSU worksheet, shown in Figure 4, uses information you provide (in the yellow columns) about the Capacity Group Names and Limits, and individual LPAR Defined Capacities for each LPAR. Based on that information, and the assumption that all LPARs are using as much capacity as they can, the sheet shows the maximum capacity that each LPAR can consume when capping is active (obviously, if capping is not active, the capacity of each LPAR is limited only by its number of logical CPs).

NOTE: Unlike the remainder of the worksheets in the tool, the information about the LPAR caps is not obtained from the zpcr file, so you must enter that information yourself if you want to see the impact of your defined caps.

After the CONFIG-zXXP and CONFIG-IFL tabs come the EXPERT worksheet. The EXPERT worksheet checks to see how the logical CP topology of each LPAR adheres to IBM's Best Practices. The list of guidelines that it checks for is listed in the LPAR Design User's Guide. The list changes over time, based on IBM's internal and customer experiences, so I won't

repeat them here. The best I can say is to load your own configuration into the tool and have the tool apply the guidelines to see how few (or how many!) exceptions you will get. An example of the EXPERT worksheet is shown in [Figure 5](#).

Figure 5 - EXPERT Worksheet

ID=Watson Walker M - LPARDesign-HD-V7-T01 Current zPCR Version-9.1a EXPERT	
	Click for EXPERT NOTES - GCI Click for EXPERT NOTES - zIIP Click for EXPERT NOTES - IFL
LPAR	Suggested Improvement Notes - GCP - Machine Type = 2964
WAWOB1	(R8-GCP) - *WARNING* - The number of VL (13) is above the IBM Best Practice See: http://www-03.ibm.com/support/techdocs/atsmastr.nsf/WebIndex/TD106388 - for this Best Practice [+] NOTE - You have 2VM and 13VL When ALL VIs will be UnParked, each VM and VL will have an entitlement of 3.57%
WAWOT0	(R5-GCP) - Due to the "at least 2 LP rule", the first VL will be always Unparked Your current Guaranteed#PP is 0.08 - you will have 2-VM with an entitlement of 4% and so 2 Active LP [+] NOTE - You have 1VM and 1VL When ALL VIs will be UnParked, each VM and VL will have an entitlement of 4%
WAWOT1	(R5-GCP) - Due to the "at least 2 LP rule", the first VL will be always Unparked Your current Guaranteed#PP is 0.32 - you will have 2-VM with an entitlement of 16.05% and so 2 Active LP [+] NOTE - You have 1VM and 1VL When ALL VIs will be UnParked, each VM and VL will have an entitlement of 16.05%
WAWOX1	(R3-GCP) - You have 1-VM with an entitlement of 88% Your current Guaranteed#PP is 0.88 - raising the Weight and removing 9-LP would give you 1-VH but with less flexibility" The New Weight should be : 12 - The current Weight is : 11 - So you must decrease another Lpar Weight by : 1 to keep Total Weight(448) constant (R8-GCP) - *WARNING* - The number of VL (9) is above the IBM Best Practice See: http://www-03.ibm.com/support/techdocs/atsmastr.nsf/WebIndex/TD106388 - for this Best Practice [+] NOTE - You have 1VM and 9VL When ALL VIs will be UnParked, each VM and VL will have an entitlement of 8.84%

In this example, you can see that it found quite a few cases where the LPAR configuration was out of line with the Best Practices guidelines. Apart from the time this saves you, I also like the way some of the exception descriptions contain links to IBM documents that describe the reasoning behind the associated guideline. At the time of writing, the tool checks the CP configuration for each LPAR against eight Best Practice guidelines, plus two additional ones for zIIPs.

As you can see in [Figure 5](#), any exceptions for an LPAR are listed beside that LPAR in the EXPERT sheet. Some of the exceptions are simply statements of fact such as “The number of VL (nn) is above the IBM Best Practice”. Others contain specific information, such as “Increase the LPAR’s weight from 11 to 12, and decrease the weight of another LPAR by the same amount”. In fact, this exception message showed up for quite a number of small test LPARs in our sample data. The full wording is:

R3-GCP) - You have 1-VM with an entitlement of 94%.

Your current Guaranteed#PP is 0.94 - raising the Weight and removing 9-LP would give you 1-VH but with less flexibility.

The New Weight should be : 12 - The current Weight is : 11 - So you must decrease another Lpar Weight by : 1 to keep Total Weight(421) constant.

I wasn't sure if Alain was recommending that the LPAR *should* be reduced to just 1 logical CP, or if he was just providing this as an observation *if* you wanted to make such a change, so I checked with him. He confirmed that it was the latter, and that, like all the observations, it should be sanity checked before making any changes. After all, *you* know your configuration better than anyone. So a recommendation that makes sense when looking at a single interval might not make sense when you know what is going to happen in that LPAR two hours later. So, take the recommendations as a suggestion that you might want to consider, rather an action that *must* be taken.

The next tab is the SYNTHESIS one. This sheet provides a nice summary of the LPARs for each processor pool (general purpose CPs, zIIPs, and IFLs). An example is shown in [Figure 6](#). You can see that it consists of three groups of cells, one group each for the general purpose CP pool, the zIIP pool, and the IFL pool.

Figure 6 - SYNTHESIS Worksheet

ID= - LPARDesign-HD-V9-T02 Current zPCR Version-9.1a - SpecCfg=YES SYNTHESIS			
HiperDispatch Effect - GCP			
	W/O HD		W/ HD
#LP (Shared Pool Only)	254		64
LP/PP ratio (Shared Pool Only)	7.06		1.78
Global Statistics			
LPAR Statistics			
#LPAR-TOTAL	23		
#LPAR w/HighShare LP (Total)	4		
#LPAR w/DED LP	0		
LP Statistics			
#HighShare LP (Total)	20		
#HighShare LP (DED)	0		
#MediumShare LP	27		
#LowShare LP	207		
HiperDispatch Effect - IFL			
	W/O HD		W/ HD
#LP (Shared Pool Only)	N/A		N/A
LP/PP ratio (Shared Pool Only)	N/A		N/A
Global Statistics			
LPAR Statistics			
#LPAR IFL TOTAL	0		
#LPAR w/HighShare LP (Total)	0		
#LPAR w/DED LP	0		
LP Statistics			
#HighShare LP (Total)	0		
#HighShare LP (DED)	0		
#MediumShare LP	0		
#LowShare LP	0		
HiperDispatch Effect - zIIP			
	W/O HD		W/ HD
#LP (Shared Pool Only)	59		43
LP/PP ratio (Shared Pool Only)	5.36		3.91
Global Statistics			
LPAR Statistics			
#LPAR with zIIP	23		
#LPAR w/HighShare LP (Total)	1		
#LPAR w/DED LP	0		
LP Statistics			
#HighShare LP (Total)	2		
#HighShare LP (DED)	0		
#MediumShare LP	25		
#LowShare LP	43		

The sheet is a convenient single source of information for things like the total number of LPARs using each pool, the total number of physical CPs in that pool, the total number of logical CPs in each pool (broken out by VH, VM, and VL CPs), and the number of LPARs that have at least one VH CP.

The next worksheet is the DASHBOARD-HD one, as shown in [Figure 7 on page 21](#). I have to admit that this is one of my favorites. It provides a graphical presentation of how many Vertical High, Medium, and Low engines of each type are in each LPAR. It also shows:

- ◆ The 'fair share' of each LPAR as a percent of the total available capacity.

Note that this field does *not* take the number of logical CPs into account - it assumes that you have defined enough logical CPs for the LPAR to get its guaranteed amount of

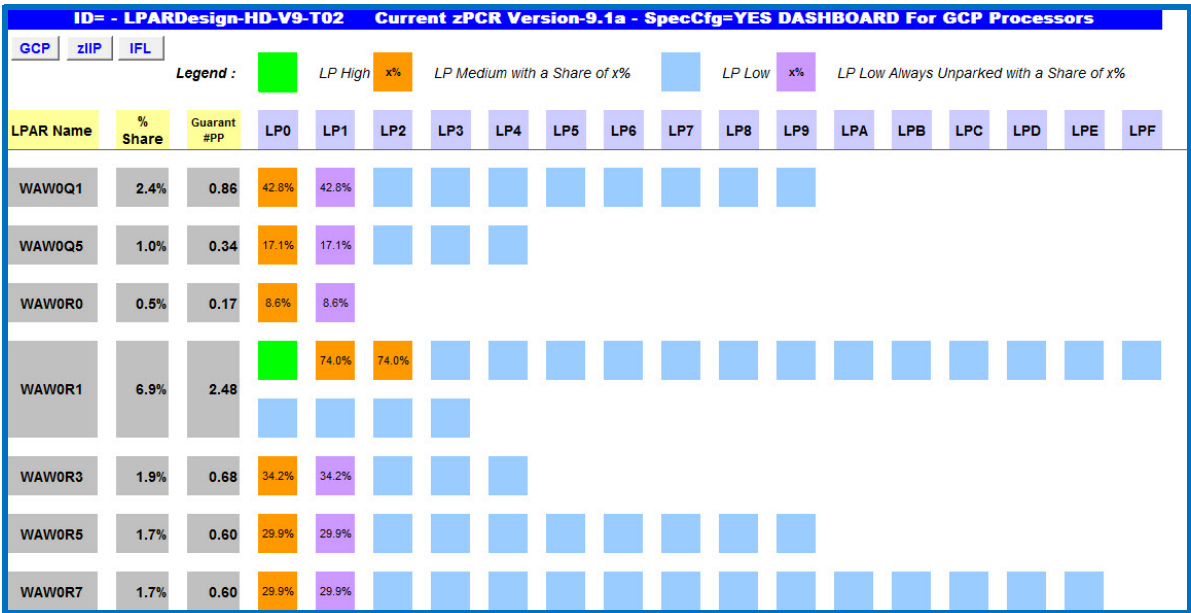
capacity. For example, if an LPAR has a weight of 50 out of 100 on a box with 100 physical CPs, but the LPAR is defined with only 5 logical CPs, then this field would show a fair share of 50% even though the effective capacity of the LPAR is really only 5%. Also, in a case like this, the capacity that that LPAR can't use (because it doesn't have enough engines) is not distributed across the other LPARs.

Such ill-defined LPARs would be identified by an OK(a) in the Check#LP column on the CONFIG worksheet. The DASHBOARD worksheet assumes that this was a mistake, and shows the fair share of each LPAR based purely on each LPAR's weight.

- ◆ The fair share in terms of how many physical CP's worth of capacity the LPAR is guaranteed.
- ◆ The percent share for each Vertical Medium CP.
- ◆ The percent share for Vertical Low CPs that are always unparked.

The different CP types (High, Medium, Low, and always-unparked-Lows) are color-coded to make it easy to see how many of each CP type are in each LPAR.

Figure 7 - DASHBOARD Worksheet



The next two worksheets, SINET and Tables, are for use by the tool and can be ignored (and should not be changed).

Using LPAR Design

Now that you have a better understanding of what the LPAR Design tool can do for you, let's have a look at how you would use it. To get the best value from this section, you should have some of your own SMF data available so you can step through the process as we go along.

RECOMMENDATION: We strongly recommend that you always pull down the latest version of the LPAR Design tool before you start a new study. Alain updates the tool regularly, with changes often reflecting hardware or PR/SM changes, so you want to make sure that you always have the most up-to-date version. If you would like to stay up to date, make sure you [subscribe to Cheryl's Blog](#)! We will post whenever an update is made to ensure you are kept in the loop.

Defining Your Configuration to LPAR Design

When you open the LPAR Design spreadsheet, you will be asked to provide a name for the study. Note - this value is only used within the spreadsheet - it does not use that value to save a copy of the spreadsheet. (It also only saves the first 15 characters of the study name, even though it allows you to enter a much longer name). Even if you have already provided a name, you will be prompted to specify the study name every time you open the tool - this allows you to give a different name for each iteration if you would like to try a few different configurations. Therefore, your next step should be to click on the big `Create a Copy` button. Give both the study and the spreadsheet a meaningful name, and one that gives you the flexibility to keep multiple iterations. And given that the spreadsheet only contains data for one CPC, include something in the file name to identify which CPC it relates to.

Having got yourself set up, the next step is to define your configuration to the tool. You can do this by entering information about your configuration on the START and CONFIG worksheets. You define the CPC on the START sheet, and the LPARs on the CONFIG sheet. If you have a small configuration, doing this manually might be the easiest way to define your configuration to the tool.

TIP: The START sheet contains a cell that says “Accept SPECIAL Conf?” This indicates whether the tool should accept configurations that do not make sense - for example, if the number of logical CPs for an LPAR is not sufficient to deliver the capacity guaranteed by the LPAR’s weight. This defaults to YES, but we recommend changing this to NO unless you have a specific reason for defining a ‘special’ configuration.

For most installations, I think that the preferred method would be to create a .zpcr file using IBM’s zPCR tool and import that file into the LPAR Design tool. The .zpcr file should be created from EDF files (which in turn were creating using IBM’s CP3KEXTR tool).

CP3KEXTR can be downloaded from

<https://www.ibm.com/support/techdocs/atsmastr.nsf/WebIndex/PRS1381>). The input to CP3KEXTR should contain at least the SMF type 70 and type 113 records^{4 5}. Alain

⁴ The LPAR Design tool doesn't use the type 113 records itself, however, it *does* extract the RNI for each LPAR, and it places that information in the spreadsheet so that it can then include that information if you export a file for use in zPCR.

⁵ Strictly speaking, you only need one EDF file per CPC because each EDF file contains enough information about the LPAR topology to let you define the configuration to the LPAR Design tool.

recommends selecting an interval when the CPC is at its maximum utilization. If you are unable to use the CP3KEXTR tool to create the EDF files, zPCR provides the ability to import an RMF CPU Activity Report directly - but remember that the CPU MF workload characterization information will be missing if you use this method.

If you decide to use a .zpcr file to define your configuration to the LPAR Design tool, go to the CONFIG worksheet and click on the zPCR icon on the top right part of the sheet, as shown in [Figure 8](#). That presents you with two options, one of which is 'Import LPARDesign Config from a zPCR Basic Study file' - click on that option.

Figure 8 - Loading Configuration from zPCR

ID=W&W Test - LPARDesign-HD-V9-T02										Current zPCR Version-9.1a - SpecCfg=YES										LPAR DEFINITION (CP) TOLERATION%=-									
CFG-LP-VALID?	NO																												
Machine-type	3906-725																												
MSU	3,644																												
Total Weight	2,415																												
Max LPAR	85																												
LPARNAME	WEIGHT	#LP	%SHARE (By Pool)	RESERVED	Guaranteed #PP	Wkld LSPR	MinReq#LP	Check#LP	HD-HIGH%	HD-MED%	HD-MED%	HD-LOW%	Active LP	Report LP															
W3906	142	2				Low																							
W014	242	3				Low-Avg																							
Z015	196	3				Average																							
W017	302	4				Average																							
ZWHITE	500	2				Average																							
S019	300	3				Average																							
W020	300	8				Average																							
W021	58	2				Average																							
W022	DED	2				Average																							
W013	60	1				Average																							
LPAR8CHR	315	8				Average																							

Configuration Validation

Delete selected LPAR

Navigate to the .zpcr file you created using zPCR and click on Open. This loads all the required information about your configuration and avoids you having to make any changes to the START or CONFIG worksheets.

After the configuration information has been loaded, click on the Configuration Validation button. This results in all the other columns on this worksheet being populated with the information for each LPAR. Some of the more interesting things to look out for on the CONFIG worksheet are:

- ◆ **LPARNAME.** The LPAR name (note that this is the LPAR name, not the system name for the system running in that LPAR). I asked Alain if he would be willing to add the system name in a future release - he suggested that I submit a Request for Enhancement, accompanied by a nice glass of red wine :).
- ◆ **Guaranteed #PP.** This is the KEY NUMBER of the HiperDispatch calculation. It is the result of computing:

$$\#Physical_Proc \times \%Share_of_the_LPAR = N.M$$

Any change to one or both of the multiply operands changes the N.M value.

- The number of physical CPs can be changed by an upgrade, by changing the number of dedicated CPs, or by facilities such as Capacity Upgrade on Demand or Capacity BackUp.
- The %Share can be changed by:

- Adding a new LPAR.
- Deleting an LPAR.
- Changing the weight of an LPAR.
- Soft capping with a positive phantom weight.

You could use this column together with the HD-MED# column to identify LPARs with 2 Vertical Medium CPs where a small change in the LPAR's weight could result in converting one of those Vertical Mediums to a Vertical High. If the number of logical CPs defined for an LPAR is insufficient to provide the guaranteed amount of capacity, this cell will be in blue for that LPAR.

- ◆ **MinReq#LP**. This is the minimum number of logical CPs required to deliver the capacity guaranteed by the LPAR's weight.
- ◆ **Check#LP**. This column contains a green 'OK', a red 'OK(a)', or '#VL>2'. The red OK(a) indicates that the number of CPs defined for the LPAR is insufficient. #VL>2 indicate that that LPAR has more than 2 VL CPs (IBM's Best Practice guideline is to have no more than 2 Vertical Low CPs in an LPAR).
- ◆ **#Active LPs**. This is the actual number of active logical CPs, taking into account that WLM will always unpark a VL CP in a 2-CP configuration with 1 VM CP and 1 VL CP.
- ◆ **#Report LPs**. This is the number of CPs as reported by RMF. It does not take into account VLs that are always unparked (RMF reports the %SHARE of a permanently unparked VL to be 0, even though, in fact, it takes half of the share from the VM).

NO COLUMN SORTING: Unfortunately, it is not possible to sort the columns in the worksheets. For example, it would be nice to be able to sort by LPAR name or the LPAR size or the number of VLs or whatever. Alain agreed to consider this for an enhancement in a future release.

Once you have your configuration information loaded into the tool, use of the tool is very intuitive. If you use the LPAR Design User's Guide, combined with information in the [What Can I Do With the LPAR Design Tool?](#) section, you should be able to navigate your way around the tool and quickly get used to applying changes you might like to make to your actual LPAR topology. The following tips might be valuable to get you started:

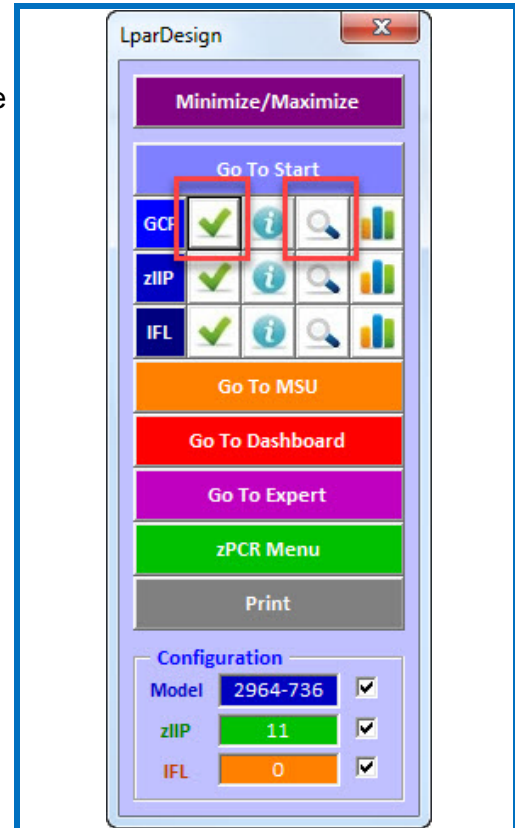
- ◆ If you close and then re-open the tool, your configuration information (including any capping information you might have entered) will still be in the sheet. However, you must click on the Configuration Validation button again (the contents of the dynamic fields are not saved when you close the spreadsheet).
- ◆ Remember that you have to click on the Configuration Validation button every time you make a set of changes (for example, if you alter the weights of a group of LPARs).

- ◆ I encountered a strange problem when using the tool. I'm not aware of anyone else that hit this same problem, but I will mention it here just in case anyone else stumbles across it. Note that this problem did not happen when I was using an external monitor - it only happens when I'm using the laptop's own screen.

When I'm using the laptop screen, when I click on the Configuration Validation buttons or any of the buttons on the EXPERT sheet, the screen flickers multiple times, then the fields are all filled in correctly, but the button roughly quadruples in size, sometimes overlaying other cells.

Figure 9 - Navigation and Action Bar

If you are not fortunate enough to have a nice external monitor for your laptop, you can avoid this problem by using the check mark icons or the magnifying glass icons in the Navigation and Action Bar, rather than using the buttons on the individual worksheets, as shown in Figure 9.



Prerequisites

The LPAR Design tool has very few prerequisites. It works with all supported levels of Microsoft Excel. It doesn't have any prerequisite z/OS level. It is intended to be used with a CPC that supports HiperDispatch, but that covers everything since z10, so hopefully that will not be a problem for anyone. While it is *recommended* to include the SMF type 113 records when creating the EDF files for the tool, they are not *required*. From an SMF perspective, the only absolute requirement is that you have type 70 records for one LPAR on the CPC that you are studying. As always, we recommend that you download the latest level of the CP3KEXTR tool from

<http://www.ibm.com/support/techdocs/atsmastr.nsf/WebIndex/PRS4229>. Each release of

the LPAR Design tool reports the newest release of zPCR that it supports. However, that does not mean that it only supports the newest release. In general, older zPCR releases are supported as well.

References

IBM is constantly improving HiperDispatch, based on their own testing experiences and the experiences of customers in the field. They have also been delivering a constant stream of mechanisms to limit the capacity available to LPARs. Therefore, it is important to keep up to date with these changes - if you are attending SHARE, watch out for sessions by **John Burg**, **Todd Havekost**, **David Hutton**, **Gary King**, or **Kathy Walsh**. In addition, the following documents provide valuable background reading:

- ◆ [HiperDispatch Questions and Answers](#) article by Alain Maneville, **Maxime Rochemir**, and Frank Kyne in *Tuning Letter 2015 No. 4*.
- ◆ *LPAR Design User's Guide*, available as part of the LPAR Design [download](#).
- ◆ SHARE in Providence Session [21045](#), *Achieving CPU (& MLC) Savings through Optimizing Processor Cache*, by Todd Havekost.
- ◆ SHARE in Providence Session [21538](#), *zPCR Capacity Sizing*, by **Brad Snyder** and John Burg. The sample files for this lab can be downloaded from the Education tab of the [zPCR Home page](#).
- ◆ SHARE in Providence Session [21572](#), *The RNI-based LSPR and the Latest z Systems Processors*, by David Hutton.
- ◆ TechDoc [106388](#), *Number of Logical CPs Defined for an LPAR*, by Kathy Wash.
- ◆ zUniversity, Munich 2017, session *CPU sizing with SMF113, LPAR Design and zPCR tools*, by Alain Maneville. This excellent presentation steps through the process of identifying the right z14 model to replace a z13 737 - an excellent guide for anyone not familiar with using zPCR. (Note - we plan to place a copy of Alain's presentation on the [Free Presentations page](#) of our website in the near future).

Summary

You might not fully appreciate the amount of work involved in determining exactly how your LPAR definitions interact with HiperDispatch until you are presented with a configuration containing a reasonable number of LPARs and physical CPs, and are asked to optimize the HiperDispatch topology. Alain's tool makes this process *so much* easier. It would not be impossible to create a spreadsheet yourself that would help with some of the calculations,

but it is unlikely that you would have all the information about the HiperDispatch logic to ensure that the results are always precise. And besides, if LPAR Design is free and easy to use, why on earth would you want to go to the trouble of re-inventing the wheel?

Alain has already implemented many enhancements since the first release of LPAR Design. And just while working with the tool to prepare for this article, I came up with a list of enhancement requests that should ensure that Alain never gets bored for the next six months. But even if he doesn't implement any of them, this is still a very valuable tool in its current form. It will certainly be a vital part of my toolkit any time I'm asked to help with optimizing a HiperDispatch configuration. I hope that you will find it as valuable as we have.